

PROPOSTA DI GRANT GIOVANI IN CSN5 - Anno 2026

Acronimo:	PRISM
Titolo completo:	Precision Room-temperature Instrumentation for high-energy x-ray absorption Spectroscopy Measurements
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Abstract (2000 caratteri)

X-ray absorption spectroscopy (XAS) is a powerful element-selective technique for probing the electronic and atomic structure of materials. While synchrotron and free-electron laser facilities provide outstanding performance, their limited accessibility motivates advanced laboratory instrumentation. A major challenge is extending fluorescence-mode XAS beyond 20 keV, where silicon detectors lose efficiency and germanium detectors require cryogenic operation. This limits studies of elements such as Mo, Rh, Ag and Cd, whose K-edges are relevant to catalysis, electrochemistry, advanced alloys and energy materials. PRISM will develop a high-resolution laboratory hard X-ray spectrometer combining advanced crystal optics with a multichannel cadmium zinc telluride (CZT) detector for precision fluorescence spectroscopy up to 50 keV. Building on my expertise in crystal-based X-ray spectroscopy gained within the VOXES project and SIDDHARTA-2 experiment, PRISM will develop detector architectures, low-noise front-end electronics, calibration procedures and spectral reconstruction algorithms to achieve sub-keV energy resolution, high quantum efficiency (>90% up to ~40 keV) and excellent SNR performance at room temperature. The instrument will be validated through fluorescence XAS measurements on reference materials with K-edges above 20 keV, demonstrating accurate edge reconstruction and XANES spectra within laboratory-compatible acquisition times. The resulting capability will enable extended in situ and operando studies of slowly evolving catalytic, electrochemical and energy-related processes. Beyond this proof of concept, PRISM will establish a new detector platform for next-generation laboratory hard X-ray spectroscopy, strengthening INFN expertise in semiconductor radiation detectors, precision X-ray instrumentation and crystal spectroscopy. The technology will complement future photon-source infrastructures such as EuAPS and support research and industrial applications.

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1 State of the art (4000 caratteri)

The performance of catalysts, electrochemical systems and advanced functional materials is governed by changes in their electronic and local atomic structure under operating conditions. These transformations are often reversible, transient or strongly dependent on the surrounding environment, and may therefore be missed by conventional ex situ characterization [1]. Understanding such processes requires in situ and operando techniques capable of probing realistic samples, including thick, heterogeneous or diluted systems, while preserving sensitivity to specific chemical elements and to their local coordination environment [2,3]. This creates a strong need for non-destructive, element-selective and bulk-sensitive probes compatible with realistic experimental environments and extended acquisition campaigns.

X-rays are particularly suited to the investigation of functional materials under realistic conditions because their high penetration depth enables access to the bulk of samples and through complex sample environments [4]. By tuning the photon energy to the absorption edge of a selected element, X-ray spectroscopies provide element- and orbital-specific information, even in chemically complex and heterogeneous systems. Moreover, they are sensitive to both electronic and local atomic structure and do not require long-range crystalline order, allowing the investigation of crystalline, amorphous, liquid and diluted materials. These properties make X-ray methods well-suited to non-destructive in situ and operando studies.

Among X-ray techniques, X-ray absorption spectroscopy (XAS) is particularly suited to this purpose because it measures the energy-dependent absorption coefficient across an element-specific absorption edge [5,6]. The near-edge region, XANES, is sensitive to oxidation state, electronic structure and coordination geometry, whereas the extended region, EXAFS, provides quantitative information on the local atomic environment, including interatomic distances, coordination numbers and structural disorder. Since XAS probes short-range order, it can be applied to crystalline, amorphous, liquid and heterogeneous materials. XAS can be measured in transmission or fluorescence mode. While transmission is generally preferred for homogeneous samples of suitable thickness, fluorescence detection is essential for diluted absorbers, thick specimens and complex sample environments, where the transmitted intensity is too weak or difficult to access.

To perform high-quality XAS measurements are conventionally performed at synchrotron facilities, whose high brilliance, broad energy tunability and advanced beamlines enable high signal-to-noise measurements over a wide energy range. XFELs further provide ultrashort and intense X-ray pulses, opening access to femtosecond time-resolved experiments and transient states. However, access to these large-scale facilities is limited and competitive, and the allocated beamtime is often unsuitable for routine characterization or experiments requiring acquisition campaigns lasting several days or weeks.

These limitations have driven renewed interest in laboratory XAS systems [7] based on conventional X-ray tubes, rotating anodes, liquid-metal-jet sources and compact accelerator-based sources. Although laboratory sources provide substantially lower brilliance than synchrotrons, they offer continuous access, flexible sample handling and long measurement times, making them attractive for routine analysis, method development and slowly evolving in situ or operando processes. Recent advances in source brightness, crystal optics and detection technology have enabled an increasing number of XAS and XES measurements to be transferred back to the laboratory.

Within INFN, VOXES [8] has established expertise in high-resolution von Hamos spectroscopy for extended and dilute X-ray sources. The instrument currently operates mainly in the 5–20 keV range

and has recently been upgraded with motorized positioning, energy-dispersive monitoring and mechanical compatibility with laboratory XAS measurements [9].

Despite this progress, laboratory fluorescence-XAS at energies above approximately 20 keV remains challenging. Silicon-based detectors progressively lose effectiveness in the hard-X-ray range, whereas high-purity germanium detectors offer excellent spectroscopic performance but require cryogenic cooling. Conversely, high-Z compound semiconductors such as CdTe and CdZnTe provide high hard-X-ray absorption efficiency and room-temperature operation, although their use requires careful control of charge transport, detector segmentation, electronic noise and spectral reconstruction.

As a consequence, the K-edges of technologically relevant elements such as Mo, Rh, Ag and Cd remain difficult to investigate by laboratory fluorescence-XAS under realistic experimental conditions. There is therefore a clear need for a compact platform combining high-resolution crystal optics with efficient, room-temperature hard-X-ray detection. Addressing this technological gap would extend laboratory XAS beyond 20 keV and establish a complementary capability between conventional laboratory instruments, synchrotron facilities and future compact photon sources.

2 Goals and Expected Outcome (2000 caratteri)

The main goal of the project is to develop and validate a laboratory platform for operando/in-situ fluorescence-XAS, optimized for realistic samples such as thick, diluted, heterogeneous specimens or systems not suitable for transmission measurements. **The platform will build on the VOXES experience in crystal optics and high-efficiency X-ray spectroscopy**, with the aim of quantitatively defining the conditions under which laboratory fluorescence-XAS can provide reliable chemical information.

The project is organized into the following specific **goals** and expected **outcomes**:

GOAL 1: Define, through **simulations**, the optimal requirements for the **X-ray source, crystal optics, sample geometry, and detection system**.

OUTCOME 1: An optimized spectrometer configuration, including quantitative specifications for the source, geometry, crystal parameters (curvature radius and thickness), and detector, capable of achieving an **acquisition time below 1 hour**, an **SNR > 20**, and an energy resolving power of **$E/\Delta E > 4000$** .

GOAL 2: Design and integrate a **fluorescence-XAS spectrometer** prototype based on crystal optics and SDD or Si-PN, compatible with realistic samples and **in-situ/operando environments**.

OUTCOME 2: By the end of the first year of the project, have a fully operational spectrometer prototype, including validated procedures for alignment, energy calibration, and control of count rate, dead time, and pile-up, achieving **positioning accuracy better than 10 μm** and **crystal angular reproducibility at the 0.5 mrad** level.

GOAL 3: Validate the methodology on benchmark samples and selected absorption edges, such as Fe, Ni, Mn or Cu K-edges, by comparing the measured spectra with reference data and simulations.

OUTCOME 3: Calibration error at X eV,

GOAL 4: **Demonstrate a measurement on a realistic case study under static and/or operando conditions and assess the transferability of the approach to compact or unconventional X-ray**

sources, including EuAPS.

OUTCOME 4: a reusable INFN platform for laboratory fluorescence-XAS, a validated dataset, documented simulation/analysis tools, and at least one publication or conference contribution.

- Rendi SMART
- Figure acronyms and focusing areas

3 Methodology (10000 caratteri)

The project is divided into four Work Packages (WPs):

WP1 - Simulation and Setup Optimization (M1-M9)

This WP involves the development of simulations of the experimental setup in order to determine the most suitable components for the design phase of WP2. Ray-tracing simulations will be performed for the crystal optics using the SHADOW4 software package, which has already been successfully employed for the validation of the VOXES setup. The X-ray source and radiation transport will be simulated using GEANT4. The main objective is to identify the optimal crystal parameters, including dimensions, type, and thickness, to achieve the best match with the characteristics of the EuAPS source.

WP2 - Setup Construction and Integration

Based on the results obtained in WP1 and within the allocated project budget, the most suitable crystals, motion stages, and Silicon Drift Detectors (SDDs) will be selected and integrated into the spectrometer design.

WP3 - Laboratory Validation at VOXES

To be defined.

WP4 - Demonstration at EuAPS

To be defined.

4 Organisation (4000 caratteri)

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	Year I				Year II			
	1-3	4-6	7-9	10-12	13-15	16-18	19-21	22-24
WORK PACKAGES (WP)								
WP1: Simulation and Optimization								
T1.1 - Requirements definition								
T1.2 - SHADOW4 ray-tracing optimization								
T1.3 - GEANT4 simulations								
T1.4 - Final geometry selection								
WP2: Instrument design and procurement								
T2.1 - Mechanical design								
T2.2 - Procurement of optics, mechanics and detector								
T2.3 - Integration								
T2.4 - Commissioning								
WP3: Assembly and validation at VOXES								
T3.1 - Energy calibration								
T3.2 - Resolution and efficiency measurements								
T3.3 - XAS/XES validation with laboratory sources								
T3.4 - Readiness review								
WP4: Experimental Demonstration								
T4.1 - Installation at EuAPS								
T4.2 - First XAS measurements								
T4.3 - Feasibility demonstration								
T4.4 - Final data analysis								
MILESTONES (MS)								
MS1. Optimized spectrometer design completed		x						
MS2. Hardware assembled and commissioned				x				
MS3. Laboratory validation completed						x		
MS4. Successful demonstration at EuAPS								x

5 Synergies with Other Activities (3000 caratteri)

EuAPS

VOXES

PANDORA

6 Impact and Possible Spillovers (3000 caratteri)

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7 Costs

-X-ray microfocus tube

-Crystals

-Axis

-Sample targets

-EXRS2028 Paris?

Year			
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CZT			

8 Risk Assessment

-Se EuAPS non parte?

9 References

- [1] Zaera, F. “*In-situ* and *operando* spectroscopies for the characterization of catalysts and of mechanisms of catalytic reactions.” *J. Catal.*, **404**, 900–910, 2021. doi: 10.1016/j.jcat.2021.08.013.
- [2] Bonn, M., Domke, K. F., Sulpizi, M., McCrory, C., and Peng, Z. “In situ and operando characterization: Advancing our understanding of functional materials.” *J. Chem. Phys.*, **162**(14), 140402, 2025. doi: 10.1063/5.0270688.
- [3] Yang, Y., Feijóo, J., Briega-Martos, V., Li, Q., Krumov, M., et al. “*Operando* methods: A new era of electrochemistry.” *Curr. Opin. Electrochem.*, **42**, 101403, 2023. doi: 10.1016/j.coelec.2023.101403.
- [4] Fehse, M., Iadecola, A., Simonelli, L., Longo, A., and Stievano, L. “The rise of X-ray spectroscopies for unveiling the functional mechanisms in batteries.” *Phys. Chem. Chem. Phys.*, **23**(41), 23445–23465, 2021. doi: 10.1039/D1CP03263A.
- [5] Bunker, G. *Introduction to XAFS: A Practical Guide to X-ray Absorption Fine Structure Spectroscopy*. Cambridge: Cambridge University Press, 2010. doi: 10.1017/CBO9780511809194.
- [6] Calvin, S. *XAFS for Everyone*. Boca Raton: CRC Press, 2013. doi: 10.1201/b14843.
- [7] Zimmermann, P., Peredkov, S., Abdala, P. M., DeBeer, S., Tromp, M., et al. “Modern X-ray spectroscopy: XAS and XES in the laboratory.” *Coord. Chem. Rev.*, **423**, 213466, 2020. doi: 10.1016/j.ccr.2020.213466.
- [8] Scordo, A., Curceanu, C., Miliucci, M., Shi, H., Sirghi, F., et al. “VOXES: a high precision X-ray spectrometer for diffused sources with HAPG crystals in the 2–20 keV range.” *J. Instrum.*, **13**(04), C04002, 2018. doi: 10.1088/1748-0221/13/04/C04002.
- [9] Manti, S. and Scordo, A. “Recent Developments of the VOXES Von Hamos X-Ray Spectrometer for Laboratory XES and XAS Studies.” *Eng. Innov.*, **19**, 23–29, 2026. doi: 10.4028/p-rQpe9Z.